

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460294.775 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1596 +/- 0.005
Transit depth (Rp/Rs)^2	2.55 +/- 0.16 [%]
Orbital Inclination (inc)	84.85 +/- 0.17
Airmass coefficient 1 (a1)	0.9832 +/- 0.0087
Airmass coefficient 2 (a2)	0.012 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	#3 - [89, 410]
Optimal Aperture	4.95
Optimal Annulus	9.85
Transit Duration (day)	0.0985 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1596 ± 0.005 vs 0.1276 (deviation 5.19σ)
Duration fit vs published	0.0985 d vs 0.1179 d (deviation 4.14σ)
Mid-time O–C	18.9 min
Depth SNR	25.9
Residual scatter	0.88 %

Light curve

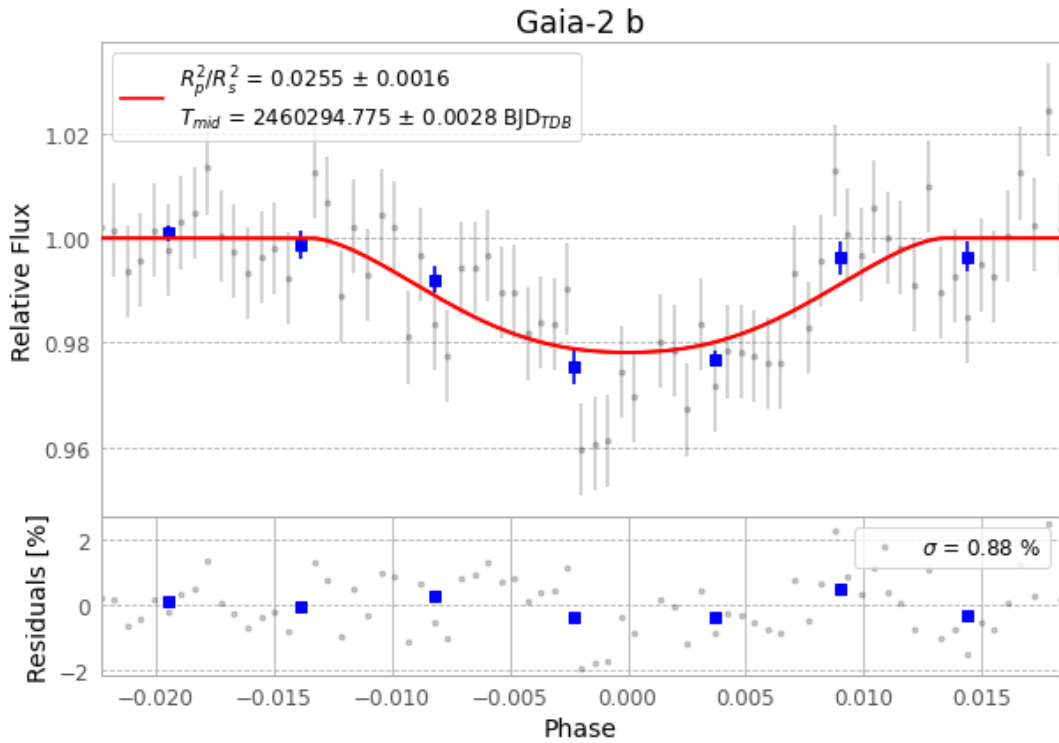


Figure 1 — FinalLightCurve_Gaia-2 b_2023-12-15.png (SHA-256 129a8bc49f1b6ae2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 efc64fd25b1a6270...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

73 FITS frames (48 MB), Gaia-2231216043015.FITS → Gaia-2231216080615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-231216.log (3dfaa95a0a4c...), FinalLightCurve_Gaia-2 b_2023-12-15.png (129a8bc49f1b...), FinalLightCurve_Gaia-2 b_2023-12-15.csv (8b602dd49b7b...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 22:48Z.